Chapter 1: Limits and Derivatives

A limit is the value that a function approaches as the input approaches some value. Formally, we say that the limit of f(x) as x approaches c is L, denoted: lim_{x -> c} f(x) = L.

The derivative of a function f at a point x represents the instantaneous rate of change. It is defined as: f'(x) = lim_{h -> 0} [f(x + h) - f(x)] / h. Key Rule: The derivative of x^n with respect to x is n * x^(n - 1).

Chapter 2: The Chain Rule

The chain rule is a formula to compute the derivative of a composite function.
If y = f(g(x)), then dy/dx = f'(g(x)) * g'(x).
Example: If f(x) = sin(x^2), the outer function is sin(u) and inner is u = x^2.
The derivative is cos(x^2) * 2x = 2x * cos(x^2).